RACoN — What Is Done, What Is Pending, What Next, and How to Make It Efficient

Evidence-grounded status report resolving the 100-question Research Dossier (27 Oct 2026) and the Research Validation / EB-2 NIW Evidence Development Plan (27 Oct 2026) against the repository docs and the live codebase. Verified 31 Oct 2026.

Method: both PDFs were text-extracted in full; all 24 markdown docs, 11 ADRs, 7 evidence files, and the measurements file were read; the codebase (racon/, protocols/, tests/, backend/, apps/, cli/) was surveyed; git log/status checked; the full test suite was executed in this session (180 passed, 3 live-Laya deselected). Status tags used: DONE (implemented + tested + evidence on disk), PARTIAL (implemented but not end-to-end proven), PENDING (planned, no implementation), UNVERIFIED (lead-reported, no independent trace).

1 — Executive verdict

Primary technical question: has a single AI model inference been partitioned and executed across multiple heterogeneous machines with verifiable per-node computation and correct end-to-end outputs? Answer: YES at 3-node scale, NOT YET at 5-node scale. CLUSTER_LAYER_SCHEDULING.md records a live split of Gemma-3-1B (26 layers 0-8 / 9-16 / 17-25) across two RPC workers plus the head with a correct returned answer, and the physical shards (451 MB + 354 MB with SHA-256 manifest) exist on disk. What is missing is the same-run-ID, five-node, per-node-compute trace with repeated runs that Gates G2/G3 demand.

Primary immigration question: can the project demonstrate Dhanasar prongs today? Answer: NO, not yet file-ready. The technical core (substantial merit vehicle) exists and is tested; the national-importance nexus (pilots, independent letters, cost model, publications) has no exhibits, and researcher-contribution exhibits (commit ledger, Scholar matrix, patent filings) are uncompiled. Sections 5-6 sequence exactly what closes each prong.

Headline contradiction resolved: STATUS.md / PRODUCTION_READINESS_AUDIT.md mark Gates 0-8 PASSED (21 Oct 2026), while MVP_ACCEPTANCE.md shows Gate 2 IN PROGRESS and IMPLEMENTATION_PLAN.md shows packaging steps pending. The audit is the newer record and matches this session's evidence (180 tests green, installer + shards on disk); the two older docs are stale and should be updated, not treated as live blockers. Second staleness: README now describes a Tauri v2 + React + FastAPI stack while older docs describe a PySide6 modular monolith — the stack migrated; architecture docs need a sync pass.

2 — Answers to all 100 dossier questions

Each answer is one verifiable statement plus its status. Evidence pointers name the exact file or observation from this session.

A. System architecture and innovation (Q1-Q8)

Q	Answer (verified 31 Oct 2026)	Status
1. Complete architecture?	Hub-and-spoke: contributor/requester PCs (outbound-only agent + UI) dial a coordinator hub behind Tailscale Funnel (worker API :8765 via Funnel, admin API :8766 loopback-only). Deterministic scheduler + optional Laya advisor, Ed25519 identity, SQLite WAL registry. See ARCHITECTURE.md topology.	DONE
2. Truly decentralized? What stays centralized?	Not fully decentralized: the coordinator hub (registry, scheduler, task dispatch) is centralized per deployment; decentralization claim rests on per-request head/temporary-coordinator behavior in cluster mode. The dossier's per-request vs fixed-coordinator wording is still unreconciled in code.	PARTIAL
3. Operate without a permanent central server?	In cluster mode the head PC runs llama-server + supervisor and replans on membership change; but coordinator services (enroll/registry/gateway keys) still assume a coordinator host. Zero-config auto-join (ADR-015) reduces operator friction but does not remove the hub.	PARTIAL
4. Node discovery + authentication?	Enrollment via /enroll with invite code or built-in join code (RCN-PUBLIC-JOIN), Ed25519 keypair per machine, SHA-256 hashed invites, PENDING/APPROVED/REVOKED trust states. Tested (coordinator+agent, abuse suites). Tailscale-IP RPC endpoint reported at enrollment.	DONE

Q	Answer (verified 31 Oct 2026)	Status
5. Requesting node as temporary coordinator?	Lead-reported design; code confirms a head/supervisor role per cluster (racon/cluster/head.py, supervisor.py), but per-request coordinator election/failover semantics are not proven with traces.	UNVERIFIED
6. Simultaneous inference requests?	Task queue + reservation locks exist; contention behavior under simultaneous requests has no measurement. No contention log.	PENDING
7. Node failure response?	Heartbeat timeout + supervisor replan: worker must fail 2 consecutive checks, then layers move to the head (which holds the full 806 MB model). In-flight requests fail once and must retry. Live kill-mid-session test not yet run.	PARTIAL
8. Difference vs existing frameworks?	Positioned as requester-led, heterogeneous consumer-hardware pipeline over llama.cpp RPC with memory-proportional planning and honest no-pooled-RAM rule — but no dated comparison matrix vs Petals/Ray/DeepSpeed/llama.cpp exists.	PENDING

B. AI model partitioning (Q9-Q20)

Q	Answer (verified 31 Oct 2026)	Status
9. Real layers / compute graphs partitioned?	Yes for Gemma-3-1B: GGUF split into 2 physical shards (layers 0-12 + embeddings; layers 13-25 + norm/head), manifest with SHA-256. Live 3-way split (0-8/9-16/17-25) returned a correct answer per CLUSTER_LAYER_SCHEDULING.md.	DONE (3-node)
10. Layer / tensor / pipeline parallelism?	Contiguous-layer pipeline parallelism over llama.cpp RPC (--rpc, -ts tensor-split fractions). Not tensor (intra-layer) parallelism.	DONE
11. Partitioning algorithm?	Memory-proportional planner (racon/cluster/planner.py): capacity_layers = floor(free_mb*0.7/block_MB), largest-remainder split, <2-layer workers dropped to head, infeasible plans rejected. Pure function + unit tests.	DONE
12. CPU / RAM / GPU / VRAM considered?	RAM/VRAM via llama-cli --list-devices --rpc measured free memory (unspoofable); CPU/GPU speed NOT yet weighted — memory-only placement is a documented limitation.	PARTIAL
13. Adaptive to heterogeneous hardware?	Share sizes scale with measured free memory per device, so yes across heterogeneous memory; no compute-speed adaptation yet.	PARTIAL
14. Weights actually distributed?	Yes: 451 MB + 354 MB shards on disk with manifest hashes; head holds full 806 MB model as fallback. HF repo toriqulhaque/RACoN-Gemma-3-1B-Shards is distribution-only, not runtime.	DONE
15. Every node executes real inference?	Verified for the 3-node live split; 5-node same-request participation unverified.	PARTIAL
16. Activation transfer?	Hidden states stream worker-to-worker over private LAN/WireGuard data plane (never Funnel). Per-job byte counts not yet instrumented.	PARTIAL
17. Computation / sync coordination?	llama.cpp RPC pipeline order + -ts fractions from the plan; head restarts llama-server on membership change. No independent barrier/timestamp trace published.	PARTIAL
18. Final result reconstruction?	Head (llama-server) aggregates pipeline output and serves via gateway; token-level correctness vs single-node baseline at fixed seed not yet published.	PENDING
19. Node without enough memory?	Planner rejects infeasible plans; <2-layer workers excluded; head fallback. True OOM-under-load test unlogged.	PARTIAL
20. Logs proving cross-node execution?	Planner/supervisor unit tests + live_plan 3-node run cited in docs; correlated 5-node run-ID bundle missing.	PARTIAL

C. AI model tests (Q21-Q31)

Q	Answer (verified 31 Oct 2026)	Status
21. Which models tested?	Gemma-3-1B sharded + live-split (verified). Dolphin family is lead-reported only; no Dolphin artifact, hash, or log found in repo.	PARTIAL

Q	Answer (verified 31 Oct 2026)	Status
22. Which Gemma 3 variant?	Gemma-3-1B-it-GLM-4.7-Flash-Heretic-Uncensored-Thinking_Q4_K_M.gguf, 806,059,360 bytes, gemma3, 26 blocks, 340 tensors, Q4_K_M, ctx 32768, emb 1152.	DONE
23. Which Dolphin release?	Not specified anywhere. No tag, base model, or hash.	PENDING
24. Sizes / formats / quantization?	Gemma: GGUF Q4_K_M as above + 2 shards with SHA-256. Dolphin: unknown.	PARTIAL
25. PCs per experiment?	Lead describes 5-machine environment; per-experiment node IDs unverified. Verified scale is 3 (head + 2 RPC workers).	UNVERIFIED
26. Did 5 PCs compute one inference?	No. Highest-priority open experiment.	PENDING
27. Per-node CPU/RAM/GPU/VRAM?	No raw utilization traces archived. Telemetry exists (psutil/Win APIs) but no per-second job traces.	PENDING
28. Machine-level proof?	Shards + manifest + planner tests + 3-node live run. No synchronized 5-node logs/screenshots/profiler bundle.	PARTIAL
29. Model one PC cannot run alone?	Proposed benefit, untested: no single-node-OOM vs distributed-success paired test.	PENDING
30. Repeatable?	Unit/proof tests repeatable (180 green here). No 3+ warm-repeat 5-node model runs.	PARTIAL
31. Latency vs single machine?	Unknown for model inference. Only control-plane RTT measured (health probe median 5.44 ms; DERP relay 15-45 ms).	PENDING

D. Performance and benchmarking (Q32-Q42)

Q	Answer (verified 31 Oct 2026)	Status
32. Tokens/sec?	No model tok/s measured. Only crypto/envelope micro-benchmarks (Ed25519 sign 0.04 ms, verify 0.11 ms; envelopes <=921 B).	PENDING
33. Time to first token?	Uninstrumented. Requires request-start to first-token-emission timing in gateway/head.	PENDING
34. 2/3/4/5-node scaling?	No scaling series. Benchmark matrix B1-B6 defined in plan; zero rows executed.	PENDING
35. Network overhead?	Qualitative only: Funnel DERP 30-80 ms makes per-token tensor streaming infeasible (ADR-012 analysis: ~1.2 s/token overhead at 10 MB). No measured per-job overhead.	PENDING
36. Model load time?	Unmeasured; docs note load time sets restart-outage length. Cold vs warm vs shard-load split needed.	PENDING
37. Memory change?	No peak per-node/aggregate comparison. Planner uses measured free memory but records no time series.	PENDING
38. Traffic bytes?	Uninstrumented per job. Envelope caps known (64 KB control, 256 KB result); tensor bytes unknown.	PENDING
39. CPU vs GPU?	No matched CPU/GPU runs. Placement ignores compute speed (documented limitation).	PENDING
40. Faster than single node?	Not established and not the right first claim: capacity expansion (run what does not fit) matters more than speed on high-latency links.	PENDING
41. Fair baseline?	Protocol defined (same model/weights/quant/prompt/tokens; note when single-node cannot load) but never executed.	PENDING
42. Run variance / CIs?	No repeats, no medians, no failure counts. Minimum: 3 warm runs per config (B1-B6).	PENDING

E. Resource-aware scheduler (Q43-Q50)

Q	Answer (verified 31 Oct 2026)	Status
43. Scheduling algorithm?	Deterministic hard-constraint filter + 4 scoring objectives (latency, throughput, efficiency, balanced); optional Laya Router typed-decisions advisor strictly among feasible IDs with deterministic fallback (ADR-002). Tested.	DONE
44. Resource monitoring?	psutil + Windows interfaces; unknown GPU reported as unavailable (no fabrication). Sampling interval not pinned in docs — needs one line in runbook.	PARTIAL
45. Refresh interval?	Heartbeat-driven (agent 10 s vs 60-120 s Funnel idle reap); staleness threshold 30 s. Exact scheduler re-poll cadence unconfirmed.	PARTIAL
46. Network cost in placement?	High-latency scenario scoring exists in Lab; exact latency/bandwidth weight in production placement formula needs a code pointer in the answer.	PARTIAL
47. Busy node?	workload_pct=100 hard-rejected; live overload injection test missing.	PARTIAL
48. Mid-run migration?	No. llama.cpp cannot hot-add devices; membership change = head restart. Treat as future work.	PENDING
49. Reproducible decisions?	Deterministic mode yes (inputs/scores/decisions logged, SQLite audit). Laya-assisted needs separate seed logging.	PARTIAL
50. Vs baselines?	No round-robin / least-loaded comparison.	PENDING

F. Reliability, networking, security (Q51-Q60)

Q	Answer (verified 31 Oct 2026)	Status
51. LAN + WAN?	LAN pipeline (RPC) implemented; WAN via Tailscale mesh/Funnel for control only. No measured LAN-vs-VPN comparison.	PARTIAL
52. Tailscale role?	Private mesh for RPC data plane + Funnel for control ingress. Mandatory only where used; LAN fallback exists.	DONE
53. Disconnect recovery?	Detect yes (heartbeat/supervisor); mid-inference recovery unproven — kill-worker test is open (Sec. 5).	PARTIAL
54. Data integrity?	SHA-256 shard manifest + signed envelopes + idempotency keys. Tensor-stream checksums/retries undocumented.	PARTIAL
55. Encryption / keys?	TLS via Tailscale + Ed25519 envelope signatures; loopback ephemeral tokens; gateway keys as SHA-256 digests. Cross-node transport audit not written.	PARTIAL
56. Unauthorized nodes rejected?	Yes: hashed single-use invites, PENDING_APPROVAL gate, revocation, tampered-signature + replay tests green.	DONE
57. Prompt/weight protection?	Weights hashed; prompts visible to executing worker operators (disclosed PUBLIC-classification notice). No further at-rest/transit doc.	PARTIAL
58. Compromised nodes?	Revocation + kill switch exist; no adversarial detection test or threat-model red-team run.	PARTIAL
59. Consent for sharing?	Privacy notice exists in SECURITY.md; installer consent screen/opt-out state unverified.	PARTIAL
60. Scale/security limits?	Known qualitative limits (Funnel throttling, restart downtime, memory-only placement); no stress-test numbers.	PENDING

G. Software and repository (Q61-Q70)

Q	Answer (verified 31 Oct 2026)	Status
61. Complete components?	Coordinator, agent, scheduler, protocols, cluster planner/supervisor/head/gateway, installer, tests — all present. Stack migrated to Tauri v2 + React + FastAPI (README) ahead of older PySide6 docs.	DONE
62. Simulated vs real?	Explicit SIM-/SIMULATED tagging enforced; scheduler Lab scenarios synthetic by design; production paths use OBSERVED/REMOTE_SINGLE_WORKER tags. One boundary audit pass still advisable.	PARTIAL
63. README matches code?	Mostly, but README (Tauri/FastAPI/SQLAlchemy) vs older docs (PySide6/modular monolith) diverged — needs sync (Sec. 6, item 1).	PARTIAL

Q	Answer (verified 31 Oct 2026)	Status
64. Test coverage?	180 passed here (31 Oct 2026) across 26 test files incl. protocols, abuse, remote-worker, cluster planner/gateway; 3 live-Laya opt-in deselected. Older docs cite 86/96 — stale counts.	DONE
65. Multi-PC e2e test?	Coordinator/agent loopback + remote-worker suites green; no 5-node model-partition e2e test in repo.	PENDING
66. Clean-Windows installer?	Inno Setup script + RACoN_Setup_v0.1.0.exe build exist (git log 17b72d5, 679bafd); fresh-VM install log missing.	PARTIAL
67. Release history?	Git history + CHANGELOG + v1.0.1 tag present; 20+ files currently modified-uncommitted — freeze and tag before evidence runs.	PARTIAL
68. Who contributed what?	No contribution ledger; commit authorship is the only record. Needs dated ledger + attestations.	PENDING
69. License / IP?	LICENSE exists; contributor assignments and ownership review with counsel not done.	PENDING
70. Outsider reproducibility?	Docs + CLI + lockfiles exist, but no independent reproduction report. Publish bundle + invite one external run.	PENDING

H. Scientific novelty and publication (Q71-Q80)

Q	Answer (verified 31 Oct 2026)	Status
71. Research problem?	Running capable AI inference on heterogeneous consumer hardware that cannot hold models alone. Needs a measurable target population.	PARTIAL
72. Literature gap?	Hypothesized only. No systematic review table.	PENDING
73. Vs Petals/Ray/DeepSpeed/llama.cpp?	No equivalent-workload comparison. Closest: ADR-012 runtime analysis (why llama.cpp RPC, not Exo/DeepSpeed/Petals) — a starting point, not a benchmark.	PENDING
74. Novel components?	Candidates: memory-proportional layer planner with unspoofable probing, zero-config join + head fallback, honest no-pooled-RAM boundary. None isolated with ablations.	PENDING
75. Borrowed vs original?	No dependency/prior-art map. llama.cpp RPC + Laya Router are borrowed; planner/supervisor/zero-config are own work — undocumented.	PENDING
76. Hypothesis?	Proposed: capacity-aware partitioning lets memory-constrained PCs run models one PC cannot hold, within acceptable overhead. Not pre-registered.	PENDING
77. Test protocol?	Defined in plan (B1-B6 + minimum record schema) but not frozen/signed before data collection.	PENDING
78. Datasets/workloads?	No standard prompt set, token limits, or license note.	PENDING
79. Negative results?	None preserved. Failure/overhead disclosure required — currently absent.	PENDING
80. Publication-ready?	No: needs reconciled architecture, replicated runs, novelty matrix. Draft only after Sec. 5.	PENDING

I. US national importance (Q81-Q90)

Q	Answer (verified 31 Oct 2026)	Status
81. US problem?	Access barriers to compute-intensive AI for labs/SMEs — plausible, undocumented with US sources.	PENDING
82. Universities/SMEs benefit?	No pilot, no measurement.	PENDING
83-84. Cost reduction + measurement?	Hypothesis only. Metric defined (TCO + cost per successful inference) but no data.	PENDING
85. Beyond one company?	Open tooling + multi-institution adoption path plausible; zero external users/citations.	PENDING
86. US sectors?	Education/research/SMB/edge hypothesized; no prioritized pilot interest.	PENDING
87. Independent interest?	No letters or adoption evidence.	PENDING

Q	Answer (verified 31 Oct 2026)	Status
88. Pilots feasible?	Possible pending DSO/counsel review (F-1). No scoped pilot.	PENDING
89-90. Roadmap / resources?	12-week plan exists; 1-3 yr deployment milestones, budget, partners, support letters all missing.	PENDING

J. Researcher and NIW evidence (Q91-Q100)

Q	Answer (verified 31 Oct 2026)	Status
91. Lead contribution?	Reported conceptual leadership; needs dated notes/commits/meeting records.	PENDING
92. Papers-to-endeavor matrix?	Publications reported but unmapped to RACoN.	PENDING
93. Citations?	Totals reported; no Scholar export or relevance mapping.	PENDING
94. Independent recognition?	Roles reported; relevance undocumented.	PENDING
95. Memberships?	IEEE etc. reported; selection-criteria relevance not shown.	PENDING
96. Patents?	Inventions reported; no RACoN filing numbers/ownership.	PENDING
97. Expert letters?	None. Needs evidence-based assessments, not generic praise.	PENDING
98-99. Experience / execution proof?	Background reported; 3-yr milestones + budget + collaborations missing.	PENDING
100. Waiver rationale?	Argument sketched; requires counsel + all Dhanasar prongs evidenced. Not a legal conclusion.	PENDING

3 — What is DONE (verified this session)

Item	Detail / evidence
Test suite green	180 passed, 3 live-Laya deselected, 31 Oct 2026, ~230 s: scheduler, protocols, coordinator/agent, abuse, remote-worker, cluster planner + gateway, storage, update manifest, chat/streaming.
Gemma-3-1B partitioned on disk	models/shards/gemma-3-1b/: 451 MB (layers 0-12) + 354 MB (layers 13-25) + shards_manifest.json with SHA-256; full 806 MB model at repo root.
3-node live pipeline run	Layers 0-8 / 9-16 / 17-25 across head + 2 RPC workers returned a correct answer (cited in CLUSTER_LAYER_SCHEDULING.md; rerun and archive the script + logs — live_plan.py is not in the repo).
Memory-proportional planner	racon/cluster/planner.py + supervisor + head + probe; largest-remainder split, <2-layer drop, infeasible-plan rejection; unit-tested.
Control/data-plane separation	ADR-007/ADR-012 enforced: Funnel carries control only (<64 KB envelopes, <256 KB results); tensors stay on private LAN/WireGuard. Empirical rationale in phase7 evaluation + funnel benchmarks.
Security baseline	Ed25519 identities, hashed invites, dual listeners (admin 404 on public port, tested), allow-listed tasks, kill switch, revocation, abuse suite green.
Zero-config join	ADR-015 implemented: RCN-PUBLIC-JOIN auto-join, per-node gateway keys, RPC auto-bind, cluster auto-discovery, autostart, re-enroll self-healing.
Packaging	PyInstaller dist + Inno Setup script + RACoN_Setup_v0.1.0.exe (v1.0.1: SSE streaming, SSD write guard, quantized KV cache, AV fix).
Gates 0-8 audit	PRODUCTION_READINESS_AUDIT.md (21 Oct 2026) marks all gates PASSED with per-gate evidence files under docs/evidence/.

4 — What is PENDING (gaps that block G2/G3/G5 and the NIW file)

Item	Detail / evidence
P1. Five-node same-run-ID trace (G2)	Highest priority. Same model/settings/prompt on 5 PCs, per-node compute spans, output correctness, 3+ warm repeats. Nothing archived yet.
P2. Benchmark matrix B1-B6 (G3)	tok/s, TTFT, load time, peak RAM/VRAM, bytes transferred, scaling 1-5 nodes, CPU-vs-GPU, baselines, CIs. Zero rows executed; instrumentation missing in gateway/head.

Item	Detail / evidence
P3. Over-memory proof	Paired test: single-node OOM vs distributed success on the same model — the core capacity claim, untested.
P4. Kill-worker mid-session test	Supervisor replan path exists; live disconnect-during-inference run with next-request-succeeds assertion missing.
P5. Dolphin evidence	Either produce tag/hash/logs or drop the claim from all materials.
P6. live_plan artifact	The cited 3-node script + logs are not in the repo; re-run under version control.
P7. Docs sync	MVP_ACCEPTANCE (Gate 2 IN PROGRESS) and IMPLEMENTATION_PLAN (packaging pending) contradict the audit; README stack (Tauri/FastAPI) vs old PySide6 docs; TESTING/STATUS counts (86/96) vs actual 180.
P8. Working-tree freeze	20+ files modified-uncommitted (incl. README, CHANGELOG, gateway, head, probe, worker). Commit/tag before any evidence run.
P9. Novelty + national-importance exhibits	No lit matrix, no prior-art benchmarks, no pilots, no letters, no cost model, no publication draft (Q71-Q90 all PENDING).
P10. Researcher exhibits	No contribution ledger, paper matrix, Scholar export, patent records, or expert letters (Q91-Q100 all PENDING); counsel review outstanding.

5 — What to do NEXT (sequenced, mapped to the 12-week plan)

Item	Detail / evidence
Week 1: freeze + reconcile (G1)	Commit/tag tree; update MVP_ACCEPTANCE + IMPLEMENTATION_PLAN to match audit; add stack-migration note; fix test counts to 180. Owner: tech lead. Closes P7/P8.
Week 2: instrument (G2/G3 setup)	Add per-job telemetry: TTFT, tok/s, bytes/node, peak RAM/VRAM, shard map, run ID in gateway/head; freeze B1-B6 protocol + prompt set + minimum-record schema. Owner: ML eng.
Weeks 3-4: 3-node rerun + 5-node trace (G2)	Re-run live 3-way split under version control (fixes P6); then 5-PC same-run-ID inference with per-node spans (fixes P1); run OOM-pair test (fixes P3). Owner: ML eng + QA.
Weeks 5-6: benchmarks (G3)	Execute B1-B6 with 3+ warm repeats, publish raw CSV + scripts + failures. Settle Dolphin (P5). Owner: QA/perf.
Weeks 7-8: reliability + security (G4)	Kill-worker test (P4), overload injection, clean-VM installer log, transport audit note, consent-screen check. Owner: systems/security lead.
Weeks 9-10: novelty + cost (G4/G5)	Related-work matrix with equivalent-workload runs; TCO + cost-per-inference model; negative-results log; paper draft skeleton. Owner: research lead.
Weeks 11-12: impact + petition (G5)	Pilot outreach (DSO/counsel-cleared), contribution ledger, Scholar export, paper matrix, expert-letter requests, counsel review, exhibit index. Owner: PI + counsel.

6 — How to make an EFFICIENT system

Efficiency for RACoN means: fit bigger models than any one PC holds, keep per-token overhead tolerable, and spend the least operator/human effort per run. The levers below are ordered by payoff and build only on mechanisms the repo already has.

Item	Detail / evidence
E1. Speed-weighted placement (biggest known gap)	Add a 60-second enrollment micro-benchmark per worker (tok/s on a calibration shard), store ms/token alongside free memory, and score capacity/speed jointly. Fixes the documented 'slow worker limits throughput' limitation without changing the planner's shape.
E2. Keep the memory-only planner honest	Retain unspoofable llama.cpp free-memory probing as the fit gate; add per-tensor head-size correction (embedding/head tensors skew the average-bytes-per-layer estimate) so plans match reality.
E3. Cut restart downtime	Membership change currently restarts llama-server (model-load seconds per flap). Batch membership updates on a short debounce; pre-stage shard loads; make clients retry idempotently (keys already exist). Track 'flap-to-serving' seconds as a KPI.
E4. Shrink per-token network cost	Keep tensors on sub-5 ms / 100+ Mbps links (already policy); add fp16 vs int8 activation option, batch small transfers, compress headers; log bytes/token so B1-B6 shows the tradeoff curve.

Item	Detail / evidence
E5. Cache aggressively	Persist KV-cache across turns (quantized KV work already started in v1.0.1); cache shard manifests + probe results per worker; reuse warm head processes across requests to amortize load time.
E6. Right-size precision	Default Q4_K_M already; document per-shard precision policy (head/gateway stays cheap, workers take what fits) and measure quality-vs-memory so 'efficient' never silently means 'wrong'.
E7. Schedule for throughput, not just fit	Extend the 4 deterministic objectives with a cost objective (expected seconds + bytes per token); compare vs round-robin/least-loaded baselines (closes Q50) and publish the win/loss.
E8. Observe everything per job	One run-ID record per inference (shard map, per-node spans, TTFT, tok/s, peaks, bytes, failures) straight into SQLite; export CSV. This single habit unblocks G2/G3, ablations (Q74), and the cost model (Q83-84).
E9. Harden the cheap paths	Rate-limit gateway before public use (quotas are a listed next step); keep Funnel control-only; keep the public join code revocable (--no-auto-join) with per-node revocation drills.
E10. Automate evidence	CI job that runs pytest, appraisal of docs-vs-code counts, shard-hash verification, and a nightly 3-node smoke split; every benchmark lands as a dated artifact. Reproducibility becomes a pipeline output, not a meeting task.

7 — Risks and claim-control rules (from the plan, endorsed)

Item	Detail / evidence
Never call task-routing 'model-parallel' without traces	Use 'partitioned' only where shard ownership + per-node compute spans exist; otherwise say routed/placed.
No speed/cost claims without controlled comparisons	Say 'potential' until B1-B6 rows exist; publish failures and variance.
Keep repo, site, and petition consistent	Audit release tags; fix unsupported website statements; do not describe Metademic as a registered nonprofit absent registration.
Protect status	F-1 authorization and pilot activity go through DSO + qualified immigration counsel before any external work.

Appendix — evidence index (this session)

Item	Detail / evidence
PDF 1 questionnaire	docs/RACoN_Project_Planning_Resent.pdf — 10 pp, 100 questions A-J, gates G1-G5, B1-B6 matrix, EV-001..006, risk register, 11-item checklist, petition draft. Fully extracted to docs/_pdf1.txt.
PDF 2 dossier	docs/RACoN_Research_Dossier_Open_This.pdf — 12 pp, per-question draft answers with PUBLIC-DOC/LEAD-REPORTED/PROPOSED/UNVERIFIED keys. Fully extracted to docs/_pdf2.txt.
Docs read	24 markdown docs + 11 ADRs + 7 evidence files + TAILSCALE_FUNNEL_BENCHMARKS.md measurement.
Code + git	racon/ (agent, cluster, coordinator, core, runtime, scheduler, storage, ui), protocols/, tests/ (26 files), backend/, apps/, cli/; HEAD dd736ec (v1.0.1); 20+ files modified-uncommitted.
Live verification	180 passed / 3 deselected (live_laya opt-in) on 31 Oct 2026; shards + manifest on disk; no live_plan.py, cluster.json, or Dolphin artifacts found in repo.